

Data Construction Guide

Monthly Stock Return Prediction Project

Final reproducibility, timing, cleaning, and validation documentation

Momen Ismail

Matriculation Number: 50307564

Institutional Email:

`s73misma@uni-bonn.de`

Machine Learning for Finance

University of Bonn

Instructor: Prof. Ivan Gufler

Public Replication Repository

<https://github.com/Momen-Ismail/stock-return-ml-project>

Final version – July 2026

Final model sample: February 1990–December 2025

Contents

How to Use This Guide	1
1 Purpose and Construction Contract	1
2 Final Verified Dataset	2
3 Pipeline Overview	3
4 Permanent and Restricted Inputs	3
4.1 Version-controlled reproducibility inputs	3
4.2 Restricted or externally acquired inputs	4
5 Critical Design Decisions	4
5.1 Locked stock universe	4
5.2 Time-aware target and predictors	4
5.3 No global preprocessing leakage	4
5.4 Monthly predictor treatment	5
5.5 Interaction design	5
6 File 01: Daily Stock Prices and Quality Cleaning	5
7 File 02: Monthly Stock Features and Target	6
7.1 Zero-volume correction	7
7.2 Market, VIX, beta, and idiosyncratic volatility	7
7.3 Target construction	8
8 File 03: Annual Compustat Characteristics and Timing	8
8.1 Accounting variables	9
9 Welch–Goyal Input Reconstruction and Validation	9
10 File 04: Raw Kelly-Style Base Panel	11
11 File 05: Imputation, Winsorization, and Interactions	11
11.1 Variable blocks	12
12 Final Predictor Groups	13
13 Timing and Leakage Controls	13
14 Validation Results	13
15 Reproduction Steps	14

15.1	Install Python requirements	14
15.2	Rebuild the Welch–Goyal input when necessary	15
15.3	Run the five data-construction stages	15
15.4	Verify the final outputs	15
16	Modeling Interface	16
16.1	Chronological samples	16
16.2	Expanding-window tuning	16
16.3	Final model set	17
16.4	Specification-selection procedure	17
16.5	Model-specific scaling	18
16.6	Test-sample rule	18
17	Relationship to the README and Main Report	18
17.1	README	18
17.2	Data Construction Guide	19
17.3	Main report	19
18	Building This Guide	19
19	Limitations	20
20	References	20

How to Use This Guide

This document is the detailed technical companion to the course-project report. The main report presents the research question, model comparison, empirical results, critical reflection, and conclusion. This guide explains exactly how the common model-ready dataset was constructed.

The guide is linked directly from the repository's main `README.md`. A reader who only wants to inspect the final empirical results does not need to read the complete document. The most relevant sections are:

- **Final Verified Dataset**, for the dimensions and final predictor architecture;
- **Pipeline Overview**, for the correct order of the five data-construction scripts;
- **Critical Design Decisions**, for the main methodological choices;
- **Timing and Leakage Controls**, for the information-availability rules;
- **Reproduction Steps**, for rebuilding the final dataset;
- **Limitations**, for the remaining data restrictions.

The detailed sections for Files 01–05 should be consulted when checking a specific variable, formula, cleaning rule, lag, merge, input, or output file.

This guide documents the construction of the common model dataset. Model estimation, hyperparameter tuning, validation-based specification selection, and final-test results are documented in the main report and in the model-output folders of the repository.

1 Purpose and Construction Contract

This document records the final construction of the monthly stock-return prediction dataset. Its purpose is to make the data pipeline reproducible, auditable, and explicit about the most important methodological decisions.

Each observation is one stock-month. Predictors are dated at month t , and the target is the stock's excess return in month $t + 1$:

$$y_{i,t+1} = r_{i,t+1} - RF_{t+1}.$$

The construction contract is:

- one row per ticker-month;
- no use of information that becomes available after the predictor month;
- a raw and unwinsorized prediction target;
- monthly cross-sectional treatment of predictor missingness and outliers;
- no model-specific standardization or sample splitting inside the common data file;
- a frozen predictor list saved next to the final dataset.

The final model-ready dataset is:

`output/data/final/006_model_dataset_kelly_winsorized_1990_2025.parquet`

The authoritative predictor list is:

`output/data/final/006_predictor_columns_kelly_winsorized.csv`

The final dataset contains:

- `ticker`;
- `month`;
- the raw target `target_excess_return_next_1m`;
- 484 predictors.

Chronological training, validation, development, and test samples are created later in the modeling workflow rather than stored as separate data-construction files.

2 Final Verified Dataset

After correcting the Welch–Goyal source and rerunning Files 03–05, the final verified dataset has:

Item	Verified value
Rows	211,059
Total columns	487
Predictors	484
Unique tickers	656
First model month	1990-02-28
Last model month	2025-12-31
Missing predictor values	0
Infinite predictor values	0
Duplicate ticker-month rows	0
Base predictors	124
Continuous stock characteristics	45
Binary stock characteristics	3
Market and VIX variables	4
Welch–Goyal macro variables	8
SIC2 dummies	64
Characteristic–macro interactions	360

The 124 base predictors are composed of:

$$45 + 3 + 4 + 8 + 64 = 124.$$

More specifically, they contain:

- 45 continuous stock characteristics;
- 3 binary stock characteristics;
- 4 market/VIX variables;
- 8 macroeconomic states;
- 64 SIC2 industry indicators.

The 360 interactions are the Cartesian product of the 45 continuous stock characteristics and the 8 macroeconomic states:

$$45 \times 8 = 360.$$

Binary characteristics, market variables, macro variables, and industry dummies are not interacted.

3 Pipeline Overview

Stage	Main script	Role
File 01	<code>src/data/01_build_clean_yaho</code> <code>o_daily.py</code>	Builds and cleans the daily Yahoo stock-price panel for the locked stock universe.
File 02	<code>src/data/02_build_monthly_st</code> <code>ock_features.py</code>	Aggregates daily observations to monthly predictors, adds market/VIX information, and constructs the next-month excess-return target.
File 03	<code>src/data/03_add_fundamentals</code> <code>_and_macro.py</code>	Cleans annual Compustat data and merges only reports considered available after a conservative six-month lag. Despite the historical filename, Welch–Goyal variables are not constructed in this file.
File 04	<code>src/data/04_build_raw_kelly_</code> <code>dataset.py</code>	Adds SIC2 dummies, merges one-month-lagged Welch–Goyal variables, validates the macroeconomic source, and freezes the 124 base predictors.
File 05	<code>src/data/05_clean_and_rank_n</code> <code>ormalize.py</code>	Performs monthly median imputation and monthly 1st/99th percentile winsorization, then constructs the 360 interactions. The filename is retained for compatibility, but the final data are not rank-normalized.

The five files must be run in numerical order. Each stage reads the output of the previous stage and saves a new auditable file.

4 Permanent and Restricted Inputs

4.1 Version-controlled reproducibility inputs

The following inputs are intended to remain under version control:

File	Role
<code>input/stock_universe_locked.csv</code>	Locked Yahoo-format stock universe used by File 01.
<code>input/market_gspc_daily.csv</code>	Permanent daily S&P 500 market series used for market returns, market volatility, beta, and idiosyncratic volatility.
<code>input/market_vix_daily.csv</code>	Permanent daily VIX series used to construct monthly VIX predictors.
<code>input/external/fama_french_rf_monthly.csv</code>	Monthly risk-free rate used to construct the excess-return target.
<code>input/external/welch_goyal_macro_1990_2025.csv</code>	Cleaned monthly macro predictors used by File 04.
<code>input/raw/PredictorData2025.xlsx</code>	Original updated Welch–Goyal workbook used to reproduce the cleaned macro CSV.
<code>input/input_manifest.csv</code>	Input registry with coverage and source notes.
<code>input/README.md</code>	Input-folder documentation.

The obsolete file:

`input/raw/PredictorData.xls`

ended in December 2005 and is not used by the final project.

4.2 Restricted or externally acquired inputs

Annual Compustat data are licensed and are not intended for public redistribution. The raw Compustat location is defined in:

`src/config.py`

A complete reconstruction therefore requires the same WRDS/Compustat extraction with the variables documented in Section 8.

The public repository contains the construction code, variable definitions, cleaning rules, six-month availability lag, and final output structure, even when the raw licensed file cannot be distributed.

Daily Yahoo stock prices are acquired or loaded by File 01 for the locked ticker universe. The locked universe and quality reports provide a reproducible record of the intended sample. However, Yahoo symbols are not permanent security identifiers, and historical vendor availability may change.

5 Critical Design Decisions

5.1 Locked stock universe

The project uses a locked S&P 500-style ticker universe rather than redownloading a changing list during each run.

This decision prevents the list of firms from changing silently across executions. It improves internal consistency and reproducibility.

However, it does not eliminate survivorship or index-selection concerns because the project does not use a fully point-in-time CRSP universe with permanent security identifiers and historical constituent membership.

5.2 Time-aware target and predictors

Predictors in month t are used to forecast excess return in month $t + 1$. Chronological ordering is enforced before all group-wise shifts and rolling calculations.

The target is:

$$\text{target_excess_return_next_1m}_{i,t} = r_{i,t+1} - RF_{t+1}.$$

The target is retained only when the next observation for the ticker is the exact next calendar month. A return spanning several calendar months is not treated as a one-month return.

The target is not winsorized in the common dataset. Large realized returns are genuine outcomes and are retained for transparent evaluation.

Any model-specific target treatment would have to be estimated only on the relevant training sample and reported separately. It is not part of the official results.

5.3 No global preprocessing leakage

The common data file does not use full-sample standardization.

Models that require scaling, including OLS-3, Elastic Net, and PLS, standardize predictors later inside training-only model pipelines.

Tree-based models use the common predictors without standardization.

Chronological validation is performed with expanding annual folds rather than random cross-validation.

5.4 Monthly predictor treatment

File 05 applies the following predictor treatment:

1. continuous stock characteristics are imputed using the same-month cross-sectional median;
2. continuous stock characteristics are winsorized at the same-month 1st and 99th percentiles;
3. binary characteristics are filled with zero when missing and remain binary;
4. market variables, macro variables, and SIC2 dummies are not rank-normalized;
5. residual missing values after the defined treatment are filled with zero;
6. the target is left unchanged.

The resulting predictors remain in economically interpretable units, subject only to monthly winsorization of the continuous stock-characteristic block.

5.5 Interaction design

The project uses a compact Kelly-style conditional-predictability design.

Only continuous stock characteristics are interacted with the eight Welch–Goyal macro states:

$$Interaction_{i,t}^{c,m} = Characteristic_{i,t}^c \times MacroState_t^m.$$

The project intentionally excludes:

- binary-characteristic $\times$ macro interactions;
- market/VIX $\times$ macro interactions;
- macro $\times$ macro interactions;
- SIC2 $\times$ macro interactions;
- interactions involving the target.

This decision keeps the feature space manageable and interpretable while still allowing characteristic effects to vary with aggregate market conditions.

6 File 01: Daily Stock Prices and Quality Cleaning

Main script:

```
src/data/01_build_clean_yahoo_daily.py
```

Purpose: create a clean daily OHLCV stock panel for the locked stock universe.

File 01 downloads or reads Yahoo data with `auto_adjust=False`, standardizes ticker and field names, sorts observations chronologically, and computes daily adjusted returns:

$$r_{i,d} = \frac{AdjClose_{i,d}}{AdjClose_{i,d-1}} - 1.$$

Daily quality rules identify invalid observations involving:

- missing core OHLCV fields;
- nonpositive adjusted prices;
- impossible OHLC relations, such as `high < low`;
- duplicate ticker-date rows;
- extreme daily returns above 300% or below -95%.

A complete ticker history is removed only when its bad-row share exceeds the configured threshold. This avoids deleting an otherwise usable history because of a small number of isolated problematic observations.

File 01 saves:

- the cleaned daily stock-price panel;
- the final clean ticker list;
- removed-ticker information;
- data-quality reports.

7 File 02: Monthly Stock Features and Target

Main script:

`src/data/02_build_monthly_stock_features.py`

Purpose: aggregate daily stock, market, VIX, and risk-free-rate inputs into a monthly stock panel. Monthly return is calculated from month-end adjusted prices:

$$r_{i,t} = \frac{AdjClose_{i,t}}{AdjClose_{i,t-1}} - 1.$$

Momentum variables use only lagged monthly returns and exclude the contemporaneous month:

$$mom_{i,t}^{(k)} = \prod_{j=1}^k (1 + r_{i,t-j}) - 1.$$

The retained monthly stock predictors include:

- recent return;
- six- and twelve-month momentum;
- long-horizon return;
- realized volatility;

- downside-return and maximum-return measures;
- dollar volume;
- Amihud illiquidity;
- zero-trading activity;
- price-trend measures;
- beta;
- idiosyncratic volatility.

Amihud illiquidity is calculated from daily data as:

$$Amihud_{i,d} = \frac{|r_{i,d}|}{AdjClose_{i,d} \times Volume_{i,d}},$$

and is then averaged within each month.

7.1 Zero-volume correction

A final audit identified 136 daily observations with invalid zero volume.

The correction was deliberately narrow. The observations were not deleted and unrelated price predictors were not altered.

Only volume-derived monthly quantities were set to missing when the underlying daily volume information was invalid:

- `avg_dolvol_1m`;
- `std_dolvol_1m`;
- `avg_log_dolvol_1m`;
- `amihud_1m`;
- `zerotrade_1m`.

The variable `dolvol_growth_1m` becomes missing automatically when its underlying monthly dollar-volume input is missing.

These missing values are later handled by the same-month median-imputation rule in File 05.

This correction preserves valid return and price information while preventing an invalid volume observation from contaminating liquidity predictors.

7.2 Market, VIX, beta, and idiosyncratic volatility

The direct market-state predictors are:

`market_ret_1m`, `market_vol_1m`, `vix_avg_1m`, `vix_change_1m`.

Rolling beta and idiosyncratic volatility use daily stock and market returns over a 252-trading-day window with a minimum of 126 observations:

$$\beta_{i,t} = \frac{\text{Cov}(r_{i,d}, r_{m,d})}{\text{Var}(r_{m,d})},$$

$$\sigma_{i,t}^{idio} = \sqrt{\max(\text{Var}(r_{i,d}) - \beta_{i,t}^2 \text{Var}(r_{m,d}), 0)}.$$

The locked risk block also includes:

$$\text{betasq_12m} = \text{beta_12m}^2.$$

7.3 Target construction

The next-month target is calculated within ticker after chronological sorting.

For a predictor row in month t , the target is the excess return realized in month $t + 1$:

$$y_{i,t+1} = r_{i,t+1} - RF_{t+1}.$$

The row is retained only when the next ticker observation is the exact next calendar month.

This rule prevents an observation after a long gap from being mislabeled as a one-month return.

8 File 03: Annual Compustat Characteristics and Timing

Main script:

```
src/data/03_add_fundamentals_and_macro.py
```

Input:

- monthly stock panel;
- raw annual Compustat file.

Outputs:

- output/data/staging/003_compustat_annual_clean.parquet;
- output/data/staging/004_monthly_panel_with_fundamentals.parquet.

Compustat `datadate` is the fiscal-year-end date, not the date on which the report became publicly available.

The project therefore imposes a conservative six-month availability lag:

$$\text{AvailableMonth} = \text{MonthEnd}(\text{datadate} + 6 \text{ months}).$$

For each ticker-month, a backward as-of merge selects the most recent Compustat observation whose availability month is not later than the stock month.

Reports older than `MAX_COMPUSTAT_AGE_MONTHS` are invalidated as stale. This prevents an old report from being carried forward indefinitely.

The latest verified File 03 results are:

- clean annual Compustat rows: 21,575;
- clean Compustat tickers: 649;
- monthly panel rows: 211,341;

- monthly panel tickers: 656;
- matched observations: 207,213;
- unmatched observations: 4,128;
- stale matches invalidated: 353;
- timing violations: 0;
- missing targets: 0.

8.1 Accounting variables

Family	Retained variables
Valuation and size	be_me, ocf_me, log_comp_market_equity
Profitability	op_at, ocf_at
Leverage and liquidity	debt_at, cash_at, cashflow_to_debt, current_ratio, quick_ratio
Investment and asset structure	capx_at, capx_growth, rd_at, ppe_at, tangibility, asset_turnover, sales_to_inventory, sales_to_cash, sales_to_receivables
Accruals, payout, and financing	accruals_at, dividend_dummy, equity_issuance_at, repurchase_at
Growth and missingness	asset_growth, sales_growth, ppeinv_gr1a, xrd_missing, has_compustat_annual
Industry	sic2 before dummy encoding

Annual growth rates are calculated only when consecutive reports are approximately one year apart. Gaps outside 300–430 days cause the affected growth variables to be set to missing rather than treating irregularly spaced reports as one-year growth.

9 Welch–Goyal Input Reconstruction and Validation

A final variable-range audit revealed that the earlier cleaned macro file contained only 124 unique values for both `wg_dp` and `wg_ep` across 431 model months.

Long exact constant sequences were present, including a 197-month block from August 2009 through December 2025.

The merge in File 04 was correct. The problem originated in the previously prepared external input. The obsolete source workbook:

`input/raw/PredictorData.xls`

ended in December 2005.

The corrected reproducibility source is:

`input/raw/PredictorData2025.xlsx`

Its monthly sheet contains 1,860 observations from January 1871 through December 2025 and includes:

- Index;
- D12;
- E12;

- `b/m`;
- `tbl`;
- `AAA`;
- `BAA`;
- `lty`;
- `ntis`;
- `svar`.

The acquisition script:

```
src/acquisition/04_create_welch_goyal_input.py
```

reconstructs the cleaned macro file for January 1990–December 2025 using:

$$wg_dp_t = \log \left(\frac{D12_t}{Index_t} \right),$$

$$wg_ep_t = \log \left(\frac{E12_t}{Index_t} \right),$$

$$wg_tms_t = lty_t - tbl_t,$$

$$wg_dfy_t = BAA_t - AAA_t.$$

The remaining variables are copied directly:

$$wg_bm = b/m, \quad wg_ntis = ntis,$$

$$wg_tbl = tbl, \quad wg_svar = svar.$$

No forward filling is used.

The validator checks:

- valid and unique monthly dates;
- a continuous monthly sequence;
- numeric macro columns without missing or infinite values;
- suspicious exact constant runs for `wg_dp` and `wg_ep`.

The corrected external file has 432 monthly observations and 432 unique values for both `wg_dp` and `wg_ep`.

After applying the one-month lag and removing January 1990, the final model panel contains 431 unique monthly values for each variable.

10 File 04: Raw Kelly-Style Base Panel

Main script:

```
src/data/04_build_raw_kelly_dataset.py
```

Inputs:

- output/data/staging/004_monthly_panel_with_fundamentals.parquet;
- input/external/welch_goyal_macro_1990_2025.csv.

Outputs:

- output/data/staging/005_raw_kelly_base.parquet;
- output/data/staging/005_raw_kelly_predictors.csv;
- output/quality/005_raw_kelly_summary.csv.

File 04 creates SIC2 industry dummies, including `sic2_missing`, and requires exactly one industry category per observation.

It then shifts each Welch–Goyal source month forward by one month before merging.

Therefore, macro information dated month t becomes a predictor for stock month $t + 1$.

January 1990 has no December 1989 macro source row in the restricted cleaned file. Its 282 stock observations therefore contain missing macro states.

This is expected. File 05 removes this complete first month, yielding a final sample beginning in February 1990.

The verified File 04 base panel contains:

- rows: 211,341;
- total columns: 127;
- tickers: 656;
- base predictors: 124;
- continuous stock characteristics: 45;
- binary stock characteristics: 3;
- market variables: 4;
- macro variables: 8;
- SIC2 dummies: 64;
- rows with missing macro states: 282, all in January 1990;
- missing targets: 0;
- infinite values: 0;
- duplicate ticker-month rows: 0.

11 File 05: Imputation, Winsorization, and Interactions

Main script:

`src/data/05_clean_and_rank_normalize.py`

Inputs:

- `output/data/staging/005_raw_kelly_base.parquet`;
- `output/data/staging/005_raw_kelly_predictors.csv`.

Outputs:

- `output/data/final/006_model_dataset_kelly_winsorized_1990_2025.parquet`;
- `output/data/final/006_predictor_columns_kelly_winsorized.csv`;
- `output/quality/006_cleaning_summary.csv`.

The script name still contains `rank_normalize` for compatibility with the earlier project structure.

The final methodology does not rank-normalize the predictors.

Instead, File 05 applies:

- monthly cross-sectional median imputation;
- monthly 1st/99th percentile winsorization of continuous stock characteristics;
- binary-value preservation;
- interaction construction;
- final data-quality checks.

11.1 Variable blocks

The 48 stock characteristics are divided into:

- 45 continuous characteristics;
- 3 binary characteristics: `dividend_dummy`, `xrd_missing`, and `has_compustat_annual`.

Continuous characteristics are imputed and winsorized within month.

Binary variables are filled with zero when missing and remain in $\{0, 1\}$.

The following blocks are preserved without cross-sectional winsorization:

- 4 market/VIX variables;
- 8 macro variables;
- 64 SIC2 dummies.

After characteristic processing, File 05 constructs 360 continuous characteristic $\times$ macro interactions.

The final file contains 484 predictors and 487 total columns.

12 Final Predictor Groups

Group	Examples and retained variables
Recent return and momentum	<code>ret_1m</code> , <code>mom6m</code> , <code>mom12m</code> , <code>long_term_return_36m</code>
Volatility and tail risk	<code>retvol_1m</code> , <code>retvol12m</code> , <code>minret_1m</code> , <code>rmax5_21d</code> , <code>beta_12m</code> , <code>betasq_12m</code> , <code>idiovol_12m</code>
Liquidity and trading activity	<code>avg_dolvol_1m</code> , <code>std_dolvol_1m</code> , <code>avg_log_dolvol_1m</code> , <code>amihud_1m</code> , <code>zerotrade_1m</code> , <code>dolvol_growth_1m</code>
Price trend	<code>price_to_ma12</code> , <code>dist_from_high_12m</code> , <code>avg_range_1m</code>
Market and VIX state	<code>market_ret_1m</code> , <code>market_vol_1m</code> , <code>vix_avg_1m</code> , <code>vix_change_1m</code>
Valuation and size	<code>be_me</code> , <code>ocf_me</code> , <code>log_comp_market_equity</code>
Profitability	<code>op_at</code> , <code>ocf_at</code>
Leverage and liquidity	<code>debt_at</code> , <code>cash_at</code> , <code>cashflow_to_debt</code> , <code>current_ratio</code> , <code>quick_ratio</code>
Investment and asset structure	<code>capx_at</code> , <code>capx_growth</code> , <code>rd_at</code> , <code>ppe_at</code> , <code>tangibility</code> , <code>asset_turnover</code> , <code>sales_to_inventory</code> , <code>sales_to_cash</code> , <code>sales_to_receivables</code>
Accruals, payout, and financing	<code>accruals_at</code> , <code>dividend_dummy</code> , <code>equity_issuance_at</code> , <code>repurchase_at</code>
Growth and missingness	<code>asset_growth</code> , <code>sales_growth</code> , <code>ppeinv_gr1a</code> , <code>xrd_missing</code> , <code>has_compustat_annual</code>
Industry controls	64 SIC2 dummies, including <code>sic2_missing</code>
Macro states	<code>wg_dp</code> , <code>wg_ep</code> , <code>wg_bm</code> , <code>wg_ntis</code> , <code>wg_tbl</code> , <code>wg_tms</code> , <code>wg_dfy</code> , <code>wg_svar</code>
Interactions	Each of the 45 continuous stock characteristics multiplied by each of the 8 macro states

13 Timing and Leakage Controls

Control	Implementation
Chronological sorting	Data are sorted by ticker and date before all shifts and rolling calculations.
Target timing	A month- t row predicts the stock's excess return in month $t + 1$.
Consecutive-month condition	The target is retained only when the next ticker observation is the exact next calendar month.
Momentum timing	Momentum variables use lagged monthly returns and exclude the contemporaneous month.
Market timing	Monthly market and VIX predictors use information observed by the end of month t .
Macro timing	Welch–Goyal month t is shifted forward and used for stock month $t + 1$.
Accounting timing	Annual Compustat information becomes available six months after fiscal-year end.
Accounting merge	A backward as-of merge prevents future reports from entering past stock months.
Stale reports	Accounting matches older than the configured maximum age are invalidated.
Imputation	Continuous characteristics use only the same month's cross-section.
Winsorization	Monthly 1st/99th percentiles are calculated only within the contemporaneous cross-section.
Model scaling	Standardization is deferred to training-only model pipelines.
Model validation	Annual expanding-window folds are used rather than random cross-validation.
Final test	The 2020–2025 test period is held out from tuning and model selection.

14 Validation Results

The final mechanical validation confirms:

- zero missing predictor values;

- zero infinite predictor values;
- zero duplicate ticker-month rows;
- zero constant predictors;
- unique predictor names;
- each SIC2 row sums to exactly one industry dummy;
- each market and macro variable has one value per month;
- all 360 interaction columns equal the exact product of their processed characteristic and macro components;
- `wg_dp` and `wg_ep` have 431 distinct values across the 431 final model months;
- the December 2025 stock rows contain November 2025 Welch–Goyal values, confirming the intended one-month lag.

The raw target is intentionally not clipped.

The variable-range audit found a minimum of approximately -0.935 and a maximum of approximately 16.251 .

These observations are retained because the target represents realized next-month excess returns.

Evaluation therefore reports MAE as a robust complement to squared-error measures, although monthly MSE remains the main project metric.

15 Reproduction Steps

All commands in this section must be run from the repository root.

The repository root is the folder containing:

- `README.md`;
- `requirements.txt`;
- `src/`;
- `input/`;
- `output/`;
- `documentation/`.

A complete reconstruction requires the licensed Compustat input. Because this file cannot be redistributed publicly, it must be placed at the location configured in:

`src/config.py`

before running File 03.

The numbered commands below are the authoritative execution order.

The five construction scripts should be run directly rather than relying on an obsolete pipeline wrapper.

15.1 Install Python requirements

From the repository root:

```
python3 -m pip install -r requirements.txt
```

A separate virtual environment is recommended.

15.2 Rebuild the Welch–Goyal input when necessary

If the cleaned Welch–Goyal CSV needs to be reconstructed from the permanent Excel source, run:

```
python3 src/acquisition/04_create_welch_goyal_input.py
```

This script is deterministic given:

```
input/raw/PredictorData2025.xlsx
```

15.3 Run the five data-construction stages

Run:

1. `python3 src/data/01_build_clean_yahoo_daily.py`
2. `python3 src/data/02_build_monthly_stock_features.py`
3. `python3 src/data/03_add_fundamentals_and_macro.py`
4. `python3 src/data/04_build_raw_kelly_dataset.py`
5. `python3 src/data/05_clean_and_rank_normalize.py`

The normal production scripts should not silently replace permanent market, VIX, Fama–French, or Welch–Goyal inputs.

15.4 Verify the final outputs

After a successful run, the main final files should exist at:

```
output/data/final/006_model_dataset_kelly_winsorized_1990_2025.parquet
```

```
output/data/final/006_predictor_columns_kelly_winsorized.csv
```

```
output/quality/006_cleaning_summary.csv
```

The final quality summary should report:

- 211,059 observations;
- 656 tickers;
- 484 predictors;
- zero missing predictor values;
- zero infinite predictor values;
- zero duplicate ticker-month rows.

16 Modeling Interface

The predictor file is authoritative.

All modeling scripts load predictors from:

```
output/data/final/006_predictor_columns_kelly_winsorized.csv
```

The predictor list should not be reconstructed manually inside individual model scripts.

This ensures that:

- every model receives the same 484 predictors;
- obsolete variables cannot enter the comparison;
- intermediate construction columns cannot be selected accidentally;
- the final model inputs remain reproducible.

16.1 Chronological samples

The final sample design is:

Sample	Period and role
Training	February 1990–December 2014. Used for fixed-model estimation and hyperparameter tuning.
Validation	January 2015–December 2019. Used to compare fixed and optimized specifications.
Development	February 1990–December 2019. Used to refit the locked final model specifications.
Test	January 2020–December 2025. Used once for official final evaluation.

The corresponding final sample sizes are:

Sample	Observations	Months	Tickers	Period
Training	131,061	299	572	Feb. 1990–Dec. 2014
Validation	35,386	60	611	Jan. 2015–Dec. 2019
Development	166,447	359	613	Feb. 1990–Dec. 2019
Test	44,612	72	641	Jan. 2020–Dec. 2025

Ticker counts differ across periods because firms enter and leave the available panel over time.

16.2 Expanding-window tuning

Hyperparameter tuning uses ten annual expanding-window folds inside the training period:

Training period	Validation year
Through December 2004	2005
Through December 2005	2006
Through December 2006	2007
Through December 2007	2008
Through December 2008	2009
Through December 2009	2010
Through December 2010	2011
Through December 2011	2012
Through December 2012	2013
Through December 2013	2014

The primary tuning criterion is average fold-level monthly mean squared error.

Random cross-validation is not used.

16.3 Final model set

The final active model set is:

- Historical Mean;
- OLS-3;
- Partial Least Squares;
- Elastic Net;
- Random Forest;
- Gradient Boosting.

A Decision Tree is not part of the official final comparison.

16.4 Specification-selection procedure

The final specifications were selected using the following procedure:

1. estimate fixed specifications on the training sample;
2. tune model families using the 2005–2014 expanding folds;
3. estimate the tuned candidates on the complete training sample;
4. compare fixed and optimized versions on the separate 2015–2019 validation period;
5. lock the final specification for each model family;
6. refit the locked models on the 1990–2019 development sample;
7. evaluate once on the untouched 2020–2025 test period.

The final validation-based choices are:

Model family	Final choice
Partial Least Squares	Optimized specification with 2 components.
Elastic Net	Optimized specification with <code>alpha</code> = 0.015 and <code>l1_ratio</code> = 0.85.
Random Forest	Fixed specification with 100 trees.
Gradient Boosting	Fixed specification with 100 trees, learning rate 0.01, and maximum depth 2.

The optimized 300-tree Random Forest and Gradient Boosting candidates were not used in the final test because they did not improve validation monthly MSE relative to the fixed 100-tree specifications.

16.5 Model-specific scaling

OLS-3, PLS, and Elastic Net use standardized predictors.

Standardization is implemented inside the model-estimation pipeline and is fitted only on the relevant training sample.

Random Forest and Gradient Boosting do not require predictor standardization because their splits depend on value ordering rather than measurement units.

16.6 Test-sample rule

The final test period must not be used for:

- hyperparameter tuning;
- predictor selection;
- choosing between fixed and optimized specifications;
- replacing one model with another;
- changing a specification after observing test performance.

The final 2020–2025 test is used once after all model decisions have been locked.

17 Relationship to the README and Main Report

The three main project documents serve different purposes.

17.1 README

The repository's main:

`README.md`

is the operational entry point.

It explains:

- where the important files are located;
- how to clone and inspect the repository;
- how to install dependencies;
- how to run the data and model stages;
- where the final outputs are stored;
- how to rebuild the report.

17.2 Data Construction Guide

This document provides the detailed technical reference for:

- input sources;
- daily and monthly filters;
- predictor definitions;
- accounting availability;
- macroeconomic timing;
- imputation;
- winsorization;
- interactions;
- validation checks;
- data limitations.

17.3 Main report

The main course-project report is stored at:

`documentation/pdf/MomenIsmail.pdf`

Its editable source is:

`documentation/documents/thesis/paper.tex`

The report presents:

- the research question;
- the model-comparison design;
- the final empirical results;
- the interpretation;
- the critical reflection;
- the conclusion.

The report should be used for the project's empirical conclusions. This guide should be used when checking how a particular observation, variable, lag, cleaning decision, or data file was constructed.

18 Building This Guide

The LaTeX source is stored at:

`documentation/documents/data_construction/Data_Construction_Guide.tex`

To compile the guide directly, run:

```
cd documentation/documents/data_construction
latexmk -pdf -interaction=nonstopmode -halt-on-error \
Data_Construction_Guide.tex
mkdir -p ../../pdf
cp Data_Construction_Guide.pdf \
../../pdf/Data_Construction_Guide.pdf
cd ../../..
```

The compiled PDF should then be available at:

`documentation/pdf/Data_Construction_Guide.pdf`

The README should link directly to this PDF so that it can be opened from the GitHub repository without compiling the LaTeX source.

19 Limitations

The final dataset is designed for a transparent course project rather than a complete replication of the CRSP/Compustat universe used in the largest empirical asset-pricing studies.

The main limitations are:

- the locked S&P 500-style universe creates survivorship and index-selection concerns;
- Yahoo ticker histories are not permanent security identifiers;
- ticker changes, mergers, acquisitions, and delistings may complicate historical matching;
- Yahoo Finance does not provide complete delisting returns;
- exact Compustat filing dates are not used;
- the six-month accounting lag is a conservative approximation rather than a report-specific publication date;
- annual rather than quarterly Compustat data limit the available accounting information;
- analyst forecasts and detailed CRSP microstructure variables are unavailable;
- some predictors are transparent Yahoo/annual-Compustat proxies rather than exact replications of every published characteristic;
- the raw return target contains large observations that can strongly affect squared-error metrics.

These limitations do not invalidate the project, but they define the scope of the empirical conclusions.

The dataset should therefore be interpreted as a transparent and leakage-controlled course-project panel rather than a complete point-in-time representation of the entire United States equity market.

20 References

References

- [1] Amihud, Y. 2002. Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*.
- [2] Ang, A., Hodrick, R., Xing, Y., and Zhang, X. 2006. The cross-section of volatility and expected returns. *Journal of Finance*.

- [3] Bali, T., Cakici, N., and Whitelaw, R. 2011. Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*.
- [4] Cboe. 2019. *VIX White Paper*.
- [5] Fama, E., and French, K. 1992. The cross-section of expected stock returns. *Journal of Finance*.
- [6] George, T., and Hwang, C. 2004. The 52-week high and momentum investing. *Journal of Finance*.
- [7] Gu, S., Kelly, B., and Xiu, D. 2020. Empirical asset pricing via machine learning. *Review of Financial Studies*.
- [8] Green, J., Hand, J., and Zhang, X. 2017. The characteristics that provide independent information about average U.S. monthly stock returns. *Review of Financial Studies*.
- [9] Jegadeesh, N., and Titman, S. 1993. Returns to buying winners and selling losers. *Journal of Finance*.
- [10] Novy-Marx, R. 2013. The other side of value: The gross profitability premium. *Journal of Financial Economics*.
- [11] Sloan, R. 1996. Do stock prices fully reflect information in accruals and cash flows? *Accounting Review*.
- [12] Titman, S., Wei, K., and Xie, F. 2004. Capital investments and stock returns. *Journal of Financial and Quantitative Analysis*.
- [13] Welch, I., and Goyal, A. 2008. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*.